

Keywords: LLM reinforcement learning • credit assignment
• value functions • privileged information

Le Critique: Privileged Value Functions for LLM Reinforcement Learning

Critics with Access to Task-Privileged Context Provide Token-Level Advantages Without Contaminating the Policy, Beating GRPO with Fewer Rollouts

By Siddarth Venkatraman, Matthieu Dinot, Laurence Aitchison

arXiv:2608.16739 • Preprint, August 2026

Group-relative methods like GRPO reduce gradient variance by sampling multiple rollouts per prompt, but they assign reward only at sequence level and are bottlenecked by straggler rollouts that reduce throughput and inflate off-policyyness. Learned value functions solve both problems in principle, but training an accurate critic from the policy’s own context is notoriously hard. *Le Critique* introduces Privileged Value Functions (PVF), which give the critic access to task-privileged information (ground-truth answer, rubric, execution state) unavailable to the policy at inference time, enabling precise token-level advantage estimates from a single rollout. An adaptive interpolation baseline (TETHER) gracefully degrades to group-relative methods when the critic is unreliable.

AT A GLANCE

Problem: GRPO provides only sequence-level credit and suffers from straggler bottlenecks; standard value functions are hard to train from the policy’s own context.

Why it matters: Token-level credit assignment could dramatically accelerate LLM RL by identifying which reasoning steps actually contribute to task success.

Method: Privileged Value Functions give critics access to task-privileged context (e.g., the ground-truth) during training only; TETHER adaptively interpolates between PVF and group baselines depending on estimated critic reliability.

Result: +4.2 accuracy points on GSM8K/MATH vs. GRPO at the same training budget; 2.3× faster convergence than PPO.

Evidence: Ablations confirm that privileged context is essential; PVF without privileged information underperforms GRPO, while TETHER provides robustness when critic quality varies.

The Big Picture

Reinforcement learning for LLMs improves model behavior on tasks with scalar reward: math problem solving, code generation, instruction following. The central challenge is efficient credit assignment: which tokens in a response contributed to earning the reward, and by how much?

Credit assignment is easy when reward is dense (every token gets a signal) or when rollouts are very short (a one-step response). For multi-step reasoning, answer construction, or long tool-use sequences, credit is sparse and long-delayed. The community has largely converged on two approaches, each with a critical flaw.

GRPO avoids a critic by constructing a group of G rollouts and using group statistics as a baseline. The flaw: every token in a response receives the same advantage, regardless of whether it was a key reasoning step or filler text.

PPO with a critic provides token-level advantages but requires training a separate model V_ϕ that can accurately predict return from the policy’s token context. The flaw: the context available to the critic (the tokens generated so far) is the same context available to the policy — and the policy’s context does not contain enough information to reliably predict long-horizon return.

Le Critique’s insight: the critic does not need to be limited to what the *policy* can see.

Why Existing Approaches Fall Short

Sequence-level baselines (GRPO, REINFORCE++): Assigning the same advantage to all tokens in a 200-token reasoning chain means the model cannot distinguish a critical deduction that led to the correct answer from a redundant step that happened to precede it. On math reasoning tasks, early reasoning steps (problem setup, variable definition) are causally most important, but they receive the same credit as conclusion steps under group-relative methods.

Standard value functions (PPO): The policy’s context at step t contains the problem statement and all tokens generated so far. This context does not directly contain the ground-truth answer, the rubric used for scoring, or whether the model is “on track” toward the correct solution. Trained on this incomplete information, critics tend to be poorly calibrated: they predict high value for high-confidence-but-wrong responses and fail to distinguish correct from incorrect intermediate reasoning.

Core Mechanism

Privileged Value Functions (PVF): The critic $V_\phi(s_t, p_t)$ receives both the standard policy context s_t (the token sequence) and a privileged context p_t available

in the training environment but not to the policy during inference. For math reasoning, p_t might include the ground-truth answer and intermediate solution steps; for code generation, it might include the test case outputs.

The privileged context is concatenated before the model’s context window at the critic input level and is never passed through the policy’s forward pass. The policy gradient uses advantages derived from the PVF:

$$\hat{A}_t = r_t + \gamma V_\phi(s_{t+1}, p_{t+1}) - V_\phi(s_t, p_t)$$

The policy parameters θ are updated via:

$$\nabla_\theta J = \mathbb{E}_t[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \hat{A}_t]$$

Gradients through the critic are blocked: ϕ is updated separately with mean squared TD error.

TETHER: Not all critics are equally reliable. TETHER adaptively interpolates between the PVF baseline and the group-relative baseline:

$$b_t = \hat{\rho} \cdot V_\phi(s_t, p_t) + (1 - \hat{\rho}) \cdot \mathbb{E}_G[R(\mathcal{T})]$$

where $\hat{\rho} \in [0, 1]$ estimates critic reliability from the held-out validation loss:

$$\hat{\rho} = \sigma(-\alpha \cdot \mathcal{L}_V^{\text{hold-out}})$$

High $\mathcal{L}_V$ (unreliable critic) drives $\hat{\rho} \rightarrow 0$, recovering group-relative training; low $\mathcal{L}_V$ (reliable critic) drives $\hat{\rho} \rightarrow 1$, using token-level PVF advantages.

Technical Formulation

Let π_θ be the LLM policy and V_ϕ the PVF critic. During a training step, one rollout $\mathcal{T} = (s_0, a_0, \dots, s_T, r_T)$ is generated from π_θ . The full TETHER policy gradient is:

$$\nabla_\theta J = \mathbb{E}_\mathcal{T} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \hat{A}_t^{\text{TETHER}} \right]$$

$$\hat{A}_t^{\text{TETHER}} = r_t + \gamma b_{t+1} - b_t$$

The critic is updated independently with:

$$\mathcal{L}_V = \mathbb{E}_t[(V_\phi(s_t, p_t) - (r_t + \gamma V_\phi(s_{t+1}, p_{t+1})))^2]$$

Learning or Inference Procedure

Training: Privileged context p_t is constructed from ground-truth task data (available in supervised datasets) and is passed to the critic at each step. The critic is updated every training iteration; reliability $\hat{\rho}$ is recomputed from a 10% held-out portion of each batch.

Inference: The critic and its privileged context are discarded entirely. The policy π_θ runs without any modifications, and privileged information is never exposed during deployment.

What the Guarantee Says

No formal guarantee is provided for PVF. The key theoretical observation is that the privileged critic, having access to the ground truth, has a much lower Bayes-optimal value estimation error than a critic trained from the policy’s context alone. This tightened value estimation translates to lower-variance advantage estimates, which in turn reduces the sample complexity of policy improvement.

Experimental Findings

- **Math reasoning (GSM8K + MATH):** PVF with TETHER achieves +4.2 accuracy points vs. GRPO at the same training budget (number of rollouts and gradient steps).
- **Training efficiency:** TETHER reaches GRPO’s final accuracy 2.3× faster in training steps, using fewer rollouts per gradient step.
- **Straggler reduction:** Eliminating the rollout-group requirement reduces training wall-clock time by 18–24% on 8-GPU configurations where straggler variance is highest.
- **PVF vs. standard critic:** PVF outperforms a standard PPO critic by +2.9 accuracy points, confirming that privileged context is the source of the advantage, not just having a critic.

Ablations and Interpretation

Privileged context content: Ground-truth answer provides most of the benefit; adding intermediate solution steps provides marginal additional gain. This suggests that knowing the target outcome is the most valuable privileged information.

TETHER vs. fixed interpolation: Fixed $\hat{\rho} = 0.5$ performs 1.8 pp worse than adaptive TETHER, confirming that critic reliability varies during training and adaptivity is needed.

Rollout group size: TETHER with $G = 1$ (single rollout) outperforms GRPO with $G = 8$ by 1.2 accuracy points, demonstrating that privileged credit assignment compensates for reduced group size.

Offline application: PVF-trained policies retain their advantage over GRPO baselines on out-of-distribution test prompts, suggesting that the improved credit assignment produces a qualitatively better-learned policy, not just in-distribution fitting.

Reference

Siddarth Venkatraman, Matthieu Dinot, Laurence Aitchison. **Le Critique: Privileged Value Functions for LLM Reinforcement Learning.** arXiv:2608.16739, Au-

gust 2026. <https://arxiv.org/abs/2608.16739>